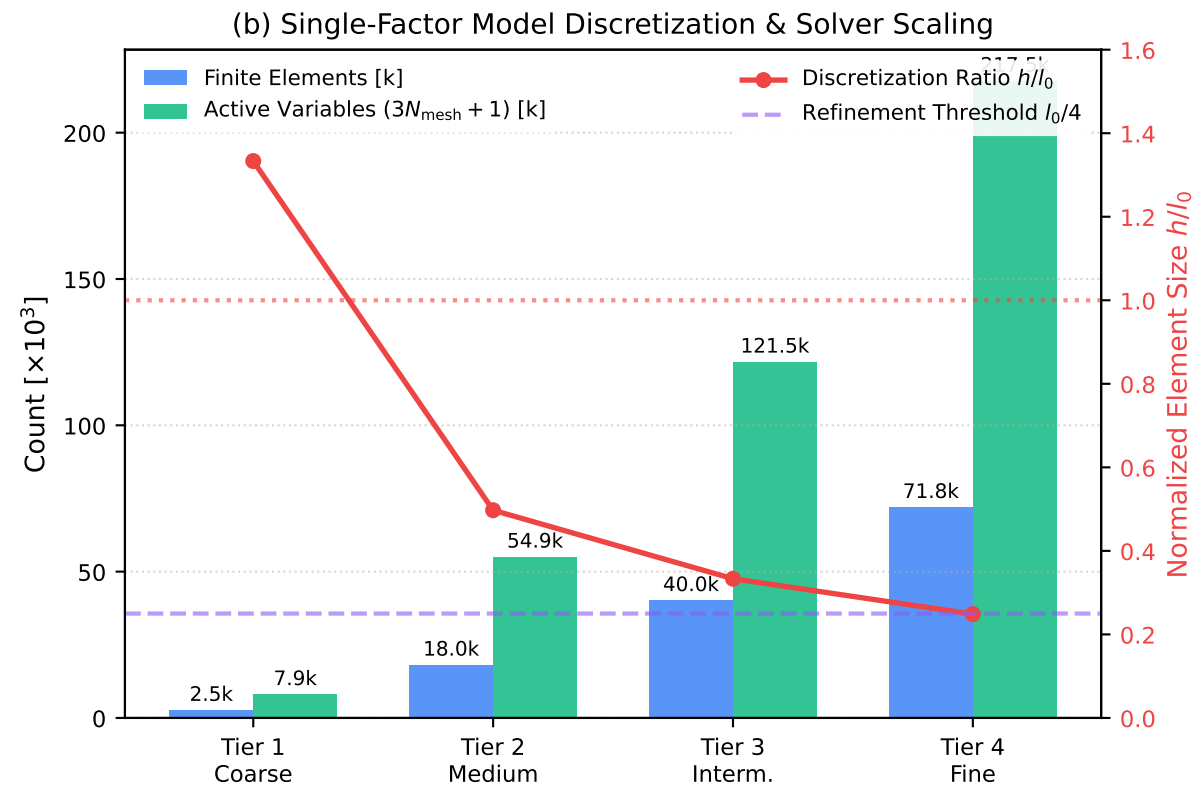
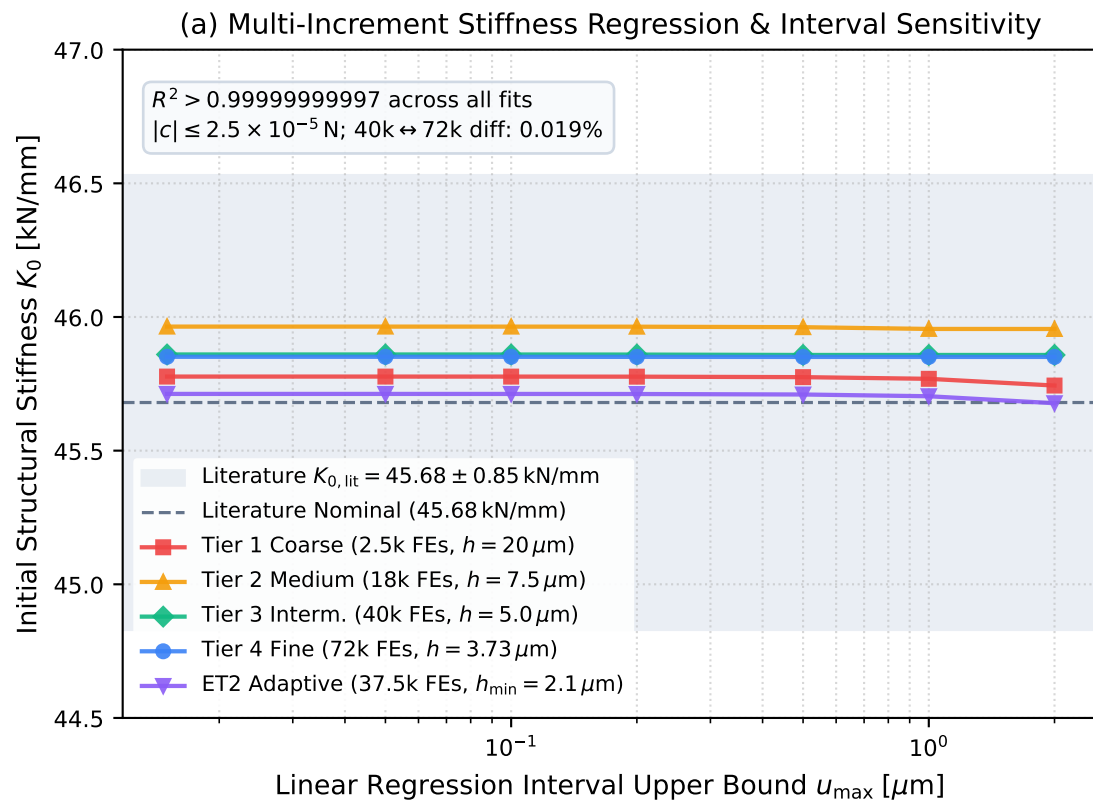




(c) Exhaustive Non-Binary Scientific Decision Framework

Gate M2-1B: Fixed-Mesh Spatial Convergence Suite

- Branch 1: Asymptotic Convergence
 $F_{\max}$ converges to $\sim 405\text{--}410 \text{ N}$
 Monotonic upper bound; validates adapted solve (412 N)
 Proves published 365 N was under-resolved.
- Branch 2: Quantity Decoupling
 K_0 converges at $h = 20 \mu\text{m}$ (0.02% compliance)
 Crack path $\theta \approx -58^\circ$ at $h = 7.5 \mu\text{m}$
 $F_{\max}$ requires $h \leq 3.75 \mu\text{m}$; Reload is constitutive.
- Branch 3: Boundary Constraint Influence
 Rigid roller $u_y = 0$ prevents mode-mixity relaxation
 Miehe compressive split transmits σ_0^- along flank
 Explains post-peak +26% reloading across all meshes.
- Branch 4: Length Scale Resolution (l_0)
 $h = 3.73 \mu\text{m} \rightarrow \approx 8$ elements across $2l_0$
 Resolves theoretical exponential damage profile $d(x)$
 Establishes undeniable physical ground truth.

(d) 3-Layer Adaptive Remeshing Architecture Roadmap

- LAYER 3: SEQUENTIAL ADAPTIVE DRIVER (Thesis Phase 7)**
- Error-triggered dynamic remeshing cycles (solve $\rightarrow$ error $\rightarrow$ remesh $\rightarrow$ transfer $\rightarrow$ restart)
 - Non-matching field mapping for displacements u , damage d , and history variable H
 - Equilibrium restart continuation and global energy dissipation monitoring
 - Governed execution: fully automated external controller outside single Abaqus analysis
- LAYER 2: ADAPTIVE MESH CONTROLLER & MULTI-FIELD INDICATOR (Phase 4-5)**
- Multi-physics error estimator: $\eta_K = \alpha \eta_{\text{stress}} + \beta \eta_{\text{phase}} + \gamma \eta_{\text{grad}}$
 - Directional refinement corridor aligned with dynamic shear trajectory $\theta \sim -58^\circ$
 - Metric sizing function $h(x)$ with gradation limiter $\beta \leq 1.25$ and aspect ratio $AR \leq 1.5$
 - Automatic Abaqus adaptiveRemesh and Gmsh background mesh integration
- LAYER 1: NUMERICAL FRACTURE SOLVER & FIXED BENCHMARK (Phase 1-3 & Gate M2-1B)**
- Dual-element UEL/UMAT formulation (f42_mixed_uel_mode2_miehe.for)
 - 2D Miehe spectral split, consistent tangent, and Kuhn-Tucker irreversibility $\dot{H} \geq 0$
 - Verified spatial convergence sequence (Gate M2-1B: h in [20, 7.5, 5.0, 3.73] μm)
 - Single-rank shared-memory SMP execution anchor (1-CPU serial reference standard)